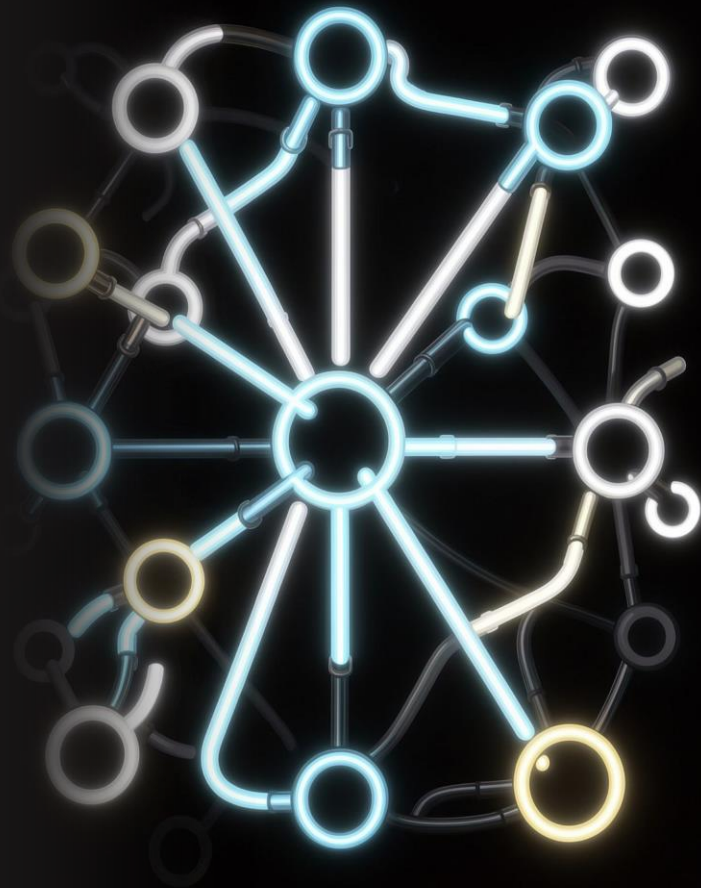


Dependency Risk Analyzer

Stop guessing. Start knowing — a data-driven approach to evaluating open-source project health.

OPEN-SOURCE INTELLIGENCE

PLATFORM RELIABILITY



The Problem: Stars Don't Equal Safety

100K Stars

Could be completely abandoned

5K Stars

Could be actively maintained

You Can't Tell

Without real signals



Four Signals That Actually Matter



Activity

Are commits happening regularly?



Team Size

One maintainer or a real team?



Response Speed

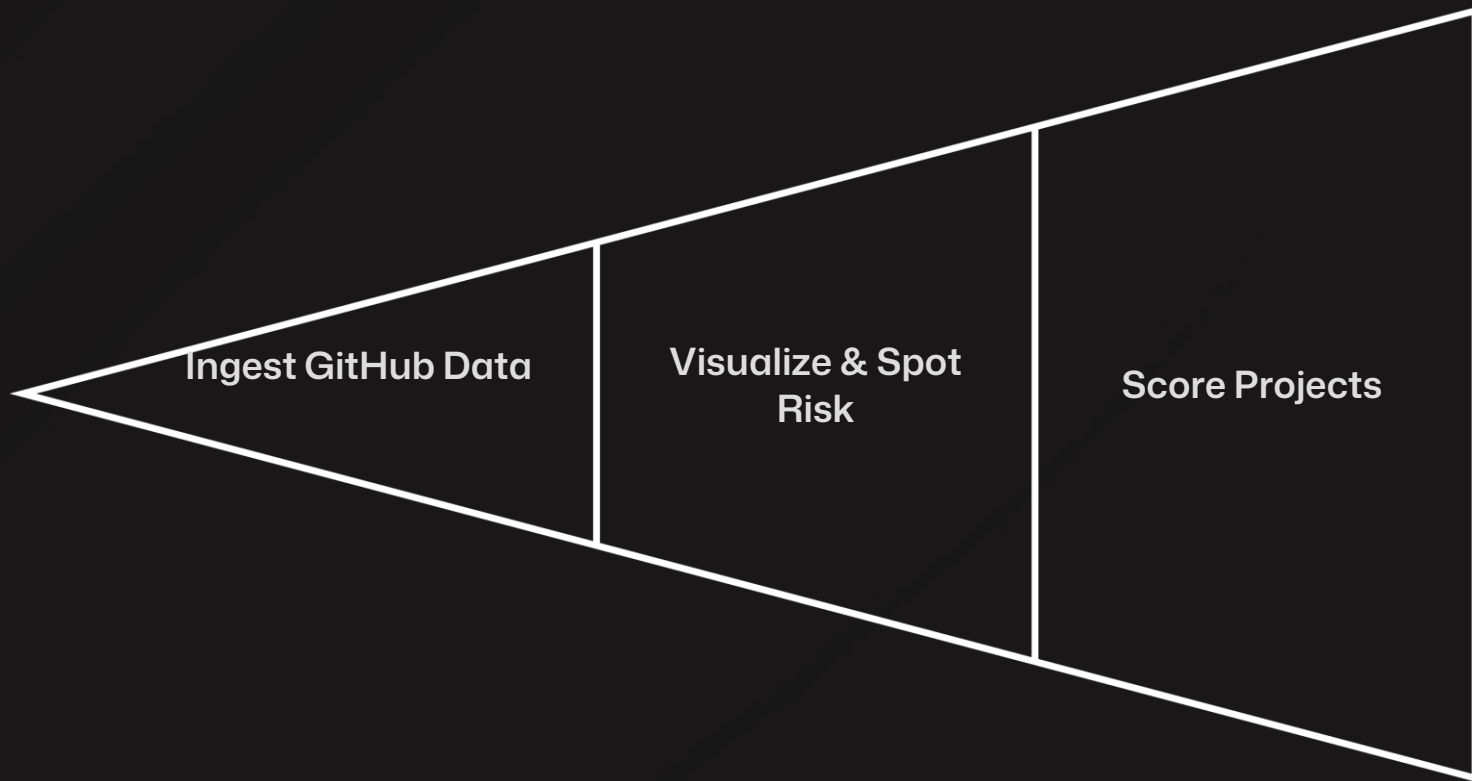
How fast are bugs fixed?



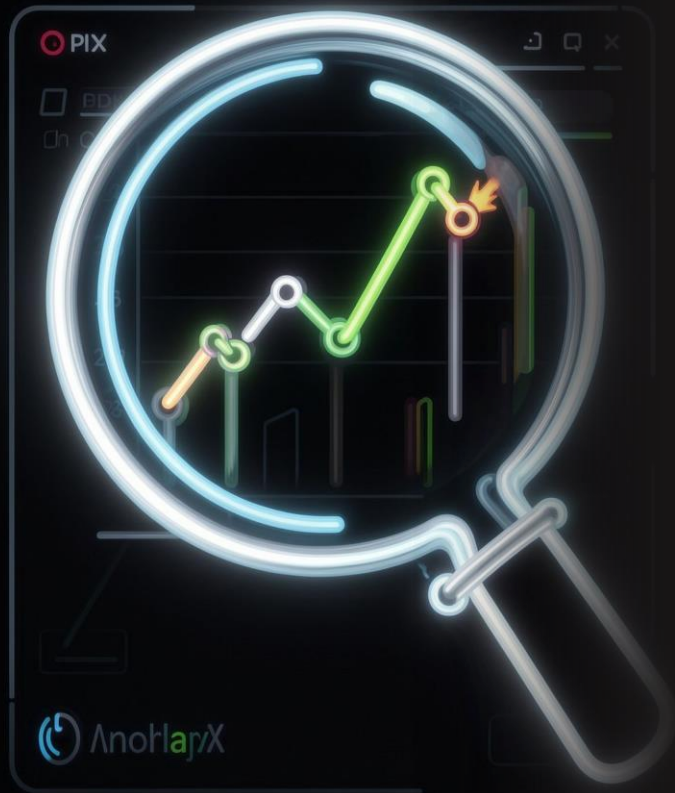
Consistency

Are updates predictable?

How It Works



Real GitHub data flows through a structured scoring engine, surfacing risk before it becomes your problem.



Key Finding: Popularity ≠ Maintenance

-0.72

Correlation

Stars vs. maintenance health —
inverse relationship

3x

Faster Closures

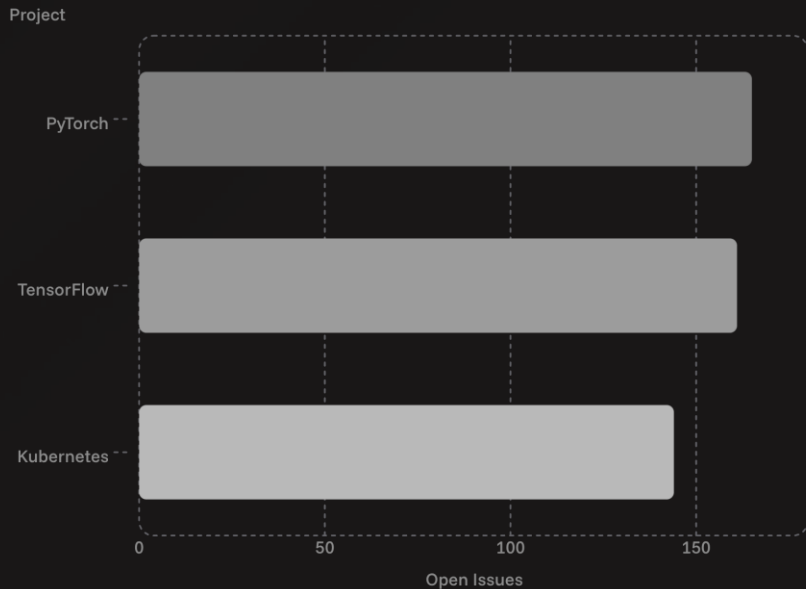
Large teams close issues 3× faster
than solo maintainers

10

Projects Analyzed

Major open-source repos evaluated

Issue Backlog Red Flags



High-backlog projects (>100 open issues)

These projects appear "healthy" on the surface — but mounting issue backlogs signal real maintenance strain despite massive adoption.

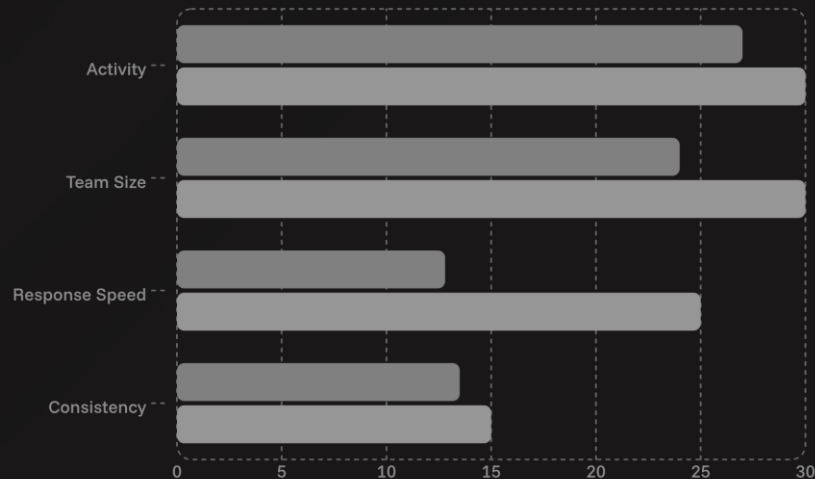


High stars + high backlog = hidden risk

Portfolio Health Breakdown

Score Max

Dimension



Overall: Mostly Healthy

8 of 10 projects marked healthy. Response Speed is the weakest link — moderate backlog across the portfolio.

✓ **Very Active**

Activity: 27/30

⚠ **Moderate Risk**

Response: 12.8/25

Tech Stack & Project Structure



SQL

DDL & data exploration scripts



Python

GitHub API fetching & health scoring



Power BI

Interactive risk visualization



Lessons Learned

1

Data Quality > Data Volume

React scored 0 due to API rate-limits — completeness matters

2

Correlation $\neq$ Causation

Stars vs. health at -0.72 partly reflects API limits, not abandonment

3

Weighting Decisions Matter

Activity 30% vs. Consistency 15% — business impact drives weights

4

API Constraints Are Real

GitHub limits 5K calls/hour — large repos need special handling

Results & What's Next

8 Healthy

Projects passing all 4 signals

2 At-Risk

100K+ stars, low maintenance

Known Limitations

- API rate-limits cap large repos at 300 commits
- 6-month window misses long-term trends
- Open-issue skew affects response scoring

Next Steps

- Store historical data for trend tracking
- Fetch more closed issues for accurate scoring